A pre-hoc test for when disproportionality nulls collapse to one, with calibrated in-regime error rates for drug–drug interaction screening: 22 years of FAERS

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August 17, 2026

Abstract

Background. Drug–drug interactions (DDIs) are largely identified after approval, and spontaneous reporting databases are the primary instrument. The established measure, Ω (Norén et al., 2008), tests the joint report count against a multiplicative null; an additive (excess-risk) null is the standard alternative. Which to prefer is usually settled by comparing how many known interactions each recovers. **We show that comparison is uninformative whenever the two nulls induce nested signal sets, give a test for when that happens that requires no outcome data, and supply the calibrated error rates the comparison should be made at instead.**

Key result. That the multiplicative assumption implies the additive one is known (Jung and Jung, 2024) but is stated unconditionally; it holds only when both marginals are elevated, since the expectations differ by (RR_A − 1)(RR_B − 1). The unconditional form fails for 163 of 16,138 generated negative controls and for none of the 6,163 with both marginals above 1. The condition is met by **98.3%** of drug-dominant pairs against **16.9%** of diffuse ones, so in the regime where DDI methods are validated the two nulls collapse to one ordering at two thresholds — checkable from the marginals before any outcome is examined.

Methods. We assembled the complete public history of the FDA Adverse Event Reporting System — 90 quarterly archives, 2004Q1–2026Q2, 328,476,258 rows — and reduced it to 20,293,421 distinct cases by six-stage deduplication including a cross-era identifier bridge validated against a chance baseline. After excluding 19,005 cases listing more than 20 drugs (see Results), the analysis population was **20,274,416 cases carrying 41,889 myotoxicity events (0.207%)**. Drug entries were resolved to active ingredients (98.0% of 73.96M rows) and the event was defined by **10 hand-curated MedDRA Preferred Terms in 3 concepts** (the `core` tier of a 23-term, 10-concept curation; the remaining 13 terms form the `broad` sensitivity tier), verified continuous across two MedDRA renamings. We validated against 16 positive controls, measured the false-positive rate against the full pool of 16,138 generated negative controls, and screened 17,375 drug pairs.

Deviation from protocol. Ω was pre-specified and **failed**, recovering 4/16 positive controls against **12/16** for an additive (excess-risk) null at the same threshold. The pooled false-positive rates of the two nulls are close (6.4% vs 6.7%), but that pooling is over a pool that does not occupy the regime where recovery is measured; in regime the rates are 2.2% (95% CI 1.24–3.34%, cluster bootstrap over drugs) and 9.3% (7.29–11.32%), so the recovery comparison is between two very different operating points (§4.3). We adopted the additive null after observing this. Ω also becomes more negative as the marginal associations strengthen (r = −0.63, n = 16, 95% CI −0.86 to −0.19, p = 0.009) — but so does Ω_add (r = −0.65, p = 0.007), while the *observed* joint event rate rises far more shallowly with marginal strength than either null predicts (r = +0.12, 95% CI −0.40 to +0.58). Both nulls therefore over-predict for strongly-associated pairs; what separates them is the size of the expectation, not its gradient. Because the estimand was selected on the evaluation set, the (binary) selection was cross-validated: the additive null wins in **16/16 leave-one-out folds**. Stable selection forces leave-one-out recovery to equal in-sample recovery, so the reported optimism of 0.000 is a consequence of that stability rather than independent evidence, and it does not address the choice of controls (§4.10). Both measures are reported throughout.

Results. Under the additive null the pipeline recovered **12/16** controls (12/14 adequately powered; 95% CI **50–100%**, resampling the victim drug, since the 16 controls are five victim drugs and simvastatin appears in seven) against 4/16 for Ω at the same threshold. Two binary design choices were available — event tier and drug role policy — and all four arms are reported: the additive advantage holds in three (+8, +8, +7 pairs) and is **null in the fourth** (broad tier with the wider role policy, 6/16 against 6/16). On a second control set selected by FDA labelling rather than by the authors (349 pairs), the same direction holds at both operating points — additive 55 vs multiplicative 29 at $\Omega_{025} > 0$, and 42 vs 15 at the calibrated threshold — but the recovery **rate falls to 12–16%**. That gap is not power: recovery *falls* as co-reporting rises, because the most heavily co-reported label-documented pairs show event rates at or below the database baseline and carry class rather than pair-specific warnings. **Neither 86% nor 12% is the sensitivity of the method**; they bracket it. The failure of Ω replicates on an independent drug-dominant event matched on event rate, torsade de pointes (**0/10** multiplicative vs 9/10 additive). The false-positive rate at $\Omega_{025} > 0$ was 6.7% against a nominal 2.5% (multiplicative 6.4%); the threshold was calibrated to Ω_add,025 > +0.436 and validated out of sample at **5.03% (95% CI 4.37–5.74%)**. The screen returned 1,022 signals of which **874 (759–997)** are expected by chance; Benjamini–Hochberg at q = 0.05 on Poisson tail probabilities returns 1,147, so the shrinkage rule is the more conservative of the two.

The screen shows no enrichment for genuine interactions once the annotation is made independent of the control set. Pooled enrichment of known interaction pairs appears significant (2.02×, 95% CI 1.59–2.55), but every positive-control drug is also on the list defining "known pair", so the annotation is not independent of the control set. Restricting to pairs containing **no** positive-control drug, enrichment is **1.12× (95% CI 0.69–1.81)** — indistinguishable from unity. This is confirmed against an **independent, endpoint-specific reference built from FDA product labelling** (709 pairs in which one drug's label names the other within 600 characters of a myotoxicity term, capturing 16/16 positive controls): enrichment 3.52× (2.71–4.56) over all pairs, falling to **1.23× (0.67–2.24)** once control drugs are excluded, and to **1.045** after stratifying on co-report count. The band in which a novel interaction would appear shows enrichment **below** unity (0.77×, 95% CI 0.66–0.89). The reference is also structurally blind to 9.9% of screened pairs — 11 of the 200 screened ingredients (5.5%) have no FDA label at all, among them fusidic acid — and restricting to pairs whose two labels both exist leaves enrichment at **1.067 (0.292–1.972)**, unchanged. Requiring the signal to persist independently in all three eras yields 19 pairs, but the same filter applied to negative controls admits 0.093% of them (95% CI 0.039–0.191%), implying **16.1 era-stable pairs by chance (95% CI 6.7–33.2) against 19 observed**; and under the endpoint-specific reference with control drugs removed, **none** of the 19 is documented. Neither how many pairs survive the filter nor which ones is distinguishable from the null.

Conclusions. We report a validated pipeline, a characterised false-positive rate, and a negative discovery result. The robust methodological finding is that the multiplicative null is unusable for events where the drugs of interest are the dominant reported cause — demonstrated on two such events. Whether that condition is what *causes* the failure remains a hypothesis: we could not assemble an adequately powered event with weak marginals to serve as the negative case. The negative discovery result is bounded by the reference rather than by the method: the screen's highest-ranked pair by event rate, atorvastatin + fusidic acid (155 events in 185 co-reports), is a contraindicated interaction that **no** reference available to us contained. A secondary finding is that 0.09% of reports — those listing more than 20 drugs — contribute 34.7% of all drug pairs at a 4× enriched event rate, and must be capped in any pairwise screen. Claims that this class of screen discovers novel interactions should be treated with scepticism until the annotation used to evaluate them is independent of the control set used to build them.

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1. Introduction

Most drug–drug interactions are discovered after approval. Pre-marketing trials cannot test the combinatorial space, so post-marketing surveillance of spontaneous reports carries the burden. FAERS is the largest public instrument for this in the United States.

Rhabdomyolysis is a natural target: severe, with well-characterised drug causes and an unusually strong positive-control set in the statin interactions. That same property — the controls being the dominant causes of the outcome — creates the methodological problem which is this paper's main contribution.

Contributions. This paper provides five things a DDI screen can use directly.

1. **The condition under which the choice of null stops mattering, and how often it is met.** That satisfying the multiplicative assumption implies satisfying the additive one is stated by Jung and Jung (2024), without conditions. It is not unconditional. With $a =$ RR_A and $b =$ RR_B the two expectations are proportional to ab and $a + b - 1$, and $ab - (a + b - 1) = (a - 1)(b - 1)$, so the implication requires both marginals on the same side of unity. We give the exact condition — **both drugs elevated** — and show it is not academic: among 16,138 generated negative controls the unconditional form fails for **163** pairs, and for **none** of the 6,163 pairs where both marginals exceed 1. The condition is met by **98.3%** of drug-dominant pairs and **16.9%** of diffuse ones, so in the regime where DDI methods are validated the two nulls are one ordering at two thresholds — and it is checkable from the marginals before any outcome is examined (§4.2).
2. **The first calibrated error rates for both nulls in the regime where DDI methods are validated**, on a purpose-built pool of 2,345 strongly-associated non-interacting pairs: $\Omega_{025} > 0$ fires on **2.2% (95% CI 1.24–3.34%)** and Ω_add,$_{025} > 0$ on **9.3% (7.29–11.32%)** against a nominal 2.5% (§4.3).
3. **A construction for regime-matched negative controls**, which the standard frequency-matched generator cannot produce, and which is reusable for any endpoint (§4.3).
4. **A one-line diagnostic for circular screen evaluation** — recompute enrichment over pairs containing no positive-control drug. Here it moves apparent enrichment from 2.02× to 1.12× (§4.5).
5. **A polypharmacy cap that improves sensitivity and false-positive rate simultaneously**, from a quantification of pair-space leverage: 0.09% of cases contribute 34.7% of all drug pairs (§4.4).

The screen itself returned no *novel* interaction, and we report that as a finding rather than burying it — but it ranked genuine contraindicated combinations first, and none of them appears in any reference available to us (§4.7).

2. Related work

Verification status. Every citation below has been verified against an indexed source; full bibliographic detail is in §8. One exception is recorded explicitly: the Ω definition used here ($\alpha = 0.5$, threshold $\Omega_{025} > 0$) is corroborated by two independent *secondary* sources — a peer-reviewed review of DDI signal-detection methods stating that "$\alpha = 0.5$

was set to provide sufficient shrinkage for avoiding disproportional highlighting based on rare reports", and the Uppsala Monitoring Centre's operational documentation listing "Interaction disproportionality measure Omega 025 > 0" — but **the primary paper is paywalled and has not been read by the authors.** A single further work on FAERS duplication practice was cited in an earlier draft and has been removed rather than cited from memory, as its bibliographic detail could not be confirmed.

Disproportionality for single drug–event pairs. The standard measures are the proportional reporting ratio (Evans, Waller and Davis, 2001), the Bayesian confidence propagation neural network / information component (Bate et al., 1998), and the multi-item gamma-Poisson shrinker (DuMouchel, 1999). All share the shrinkage logic used here: a lower credibility bound withholds a signal at low counts.

Extension to drug pairs. Thakrar, Grundschober and Doessegger (2007) set out additive and multiplicative models for interaction signals in spontaneous reports and noted that the two answer different questions. Norén, Sundberg, Bate and Edwards (2008) introduced Ω, the measure this study pre-specified, comparing the observed triple count against a log-linear model with all pairwise associations and no three-way term. Our contribution is not a new estimator but an empirical demonstration of where Ω's null breaks down, and the identification of the condition under which it does.

Additive versus multiplicative interaction. The argument that departure from *additivity* is the criterion for interaction of public-health relevance, while departure from multiplicativity answers a different question, is long-standing in epidemiology (Rothman, 1976), where departure from additivity is the test for synergism under the sufficient-cause framework. VanderWeele and Knol (2014) give the modern treatment, and their recommendation — that both scales be reported rather than one chosen — is the practice adopted here. We rediscovered this argument empirically and then found it already established; we claim novelty only for the demonstration in the spontaneous-reporting setting.

FAERS curation. Banda et al. (2016) published AEOLUS, a curated and standardised FAERS resource covering January 2004 to June 2015, retaining 4,928,413 unique cases. Over the identical window this pipeline retains 5,337,888 — a difference of 8%, in the direction expected given that AEOLUS applies an additional pass deduplicating on event date, age, sex and country alone regardless of case number (§6). The DiAna dictionary (Fusaroli et al., 2024) provides a curated FAERS drugname→ingredient mapping. We did not use it, having found that FDA's own `prod_ai` field applied backwards suffices — and the two agree closely on coverage: DiAna reports standardising **98.94%** of 74,143,411 drug entries; this pipeline resolves **98.0%** of 73,960,283. That near-agreement, reached by independent means, is the closest thing to an external check on the ingredient resolution available here.

DDI reference sets. TWOSIDES (Tatonetti et al., 2012) is derived from FAERS itself and is therefore unsuitable as an independent reference here. DrugBank is licensed and was not available.

The ONC high-priority DDI list (Phansalkar et al., 2012) *is* available — it is a 15-row table in an open-access paper, and an earlier draft of this manuscript described it as unavailable without checking. It was retrieved and examined. It is **not usable as a reference for this study**, for a reason specific to it rather than to access: the list is expressed as *drug-class* pairs (for example "HMG Co-A reductase inhibitors ↔ CYP3A4 inhibitors", "QT prolonging agents ↔ QT prolonging agents"), not as ingredient pairs, so applying it requires an ingredient→class mapping that would itself have to be author-written — the circularity this study is trying to escape. Only one of the 15 concerns myotoxicity, and the paper explicitly *excluded* the gemfibrozil–statin interaction, one of this study's positive controls, on the grounds that "clinical benefit of co-prescribing outweighs risk". A 15-row class-level list that excludes the canonical myotoxicity pair cannot serve as the endpoint-specific reference this evaluation needs.

A curated, ingredient-level, severity-graded DDI compendium remains unavailable, and that is the single largest limitation of the evaluation (§6).

3. Methods

3.1 Data acquisition

All 90 quarterly archives (3.55 GB compressed, 20.04 GB uncompressed) were downloaded with a manifest recording a SHA-256 per archive, since FDA silently re-issues quarterly files.

The archives are not one format. An audit of every column header in every table across all 90 quarters found 21 schema changepoints, three of which would have corrupted results without raising an error:

- **A UTF-8 byte-order mark** on the first column name of `DRUG12Q4.txt` and nowhere else, so every drug-to-demographics join for that quarter matches zero rows.
- **2012Q4 spells three columns differently from both neighbours** (`outc_code`/`outc_cod`, `lot_nbr`/`lot_num`, `i_f_code`/`i_f_cod`).
- `DEMO18Q1_new.txt` — a non-standard filename that a pattern anchored on `Q<digit>.TXT` classifies as documentation, silently dropping a quarter of demographics.

Deleted-case lists ship for 2019Q1–2026Q2 under five naming conventions, including `Deleted/DELETEnnQn.txt` from 2021Q4, which contains no occurrence of "deleted". A cumulative list in 2019Q1 covers everything prior.

3.2 Parsing

Every data line in every FAERS table ends with a trailing delimiter its header does not declare. Given more fields than names, pandas promotes the surplus leading column to the index and shifts every column left by one, silently. In `DRUG` this moves `drug_seq` into `primaryid`; both are integers, every type check passes, and the pipeline completes with every drug attributed to the wrong report. Parsing is therefore validated by referential integrity rather than by type: **0 orphans across all 303,663,833 child-table rows** — every `DRUG`, `REAC`, `INDI`, `THER`, `OUTC` and `RPSR` row resolves to a `DEMO` row in its own quarter. `DEMO` itself is the parent of that relation and so cannot be orphaned; it is checked differently, against the download manifest, as are the other six: parsed row counts match the manifest for all seven tables, **328,476,258 rows** in total.

3.3 Deduplication

Six stages, 24,812,425 raw demographics rows → 20,293,421 distinct cases. **The first two rows are per-era, not cumulative**: LAERS and FAERS use disjoint identifier spaces and are deduplicated separately before being bridged, so their `remaining` values are counts within their own era (4,276,201 and 20,536,224 raw rows respectively) and sum to 20,775,025 entering the bridge. Rows from the cross-era bridge down are cumulative. Seven LAERS rows carry a NULL `case_id` and are dropped before stage 1, so the two era totals sum to 24,812,418 rather than the 24,812,425 raw.

stage	remaining	removed
raw DEMO rows (all eras)	24,812,425	—
within-LAERS, per era (highest `isr` per case)	3,091,161	1,185,033
within-FAERS, per era (highest `caseversion`)	17,683,864	2,852,360
cross-era bridge	20,692,683	82,342
FDA-deleted cases	20,588,497	104,186
near-duplicates	20,294,190	294,307
one case per report	20,293,421	769

Two counts of the withdrawn cases, and why they differ. FDA publishes 229,233 withdrawn case identifiers. **104,186** of them occur in `DEMO` and are removed at stage 4; **98,102** occur in FAERS-era `DEMO` specifically. The 6,084-case difference is identifiers that appear *only* under LAERS numbering — withdrawn FAERS `caseid` values that also occur as LAERS `case` numbers. That is not a discrepancy to explain away but a second line of evidence for the shared identifier space stage 3 depends on, arrived at independently of the demographic agreement reported below. Both counts are in `results/tables/parse_validation.csv`; the removal operates on the union.

The cross-era bridge was validated, not assumed. 82,342 identifiers appear in both numbering systems. Event date agreed on 33.3% of them against 0.0% at a chance baseline; all of date/sex/age on 26.0% against 0.0%.

Near-duplicate eligibility requires event date, age, drug set and PT set rather than a count of populated fields; an initial "any 4 of 6" rule removed 14.5% of all cases with a largest group of 9,270, a collision among sparse records.

3.4 Ingredient resolution

`prod_ai` exists only from 2014Q3 (97.8% coverage after, 0% before). The modern era was used as an FDA-curated `drugname` → `ingredient` lookup and applied backwards, resolving 90.1% of LAERS rows; a relaxed pass stripping dose and packaging detail added ~6 points, for **98.0% of 73,960,283 rows**. Salt and hydrate forms are stripped; element-headed compounds (calcium carbonate, ferrous sulfate) are protected.

Coverage is not accuracy. No manual validation of resolution correctness was performed; see §6.

3.5 Event definition

23 PTs across 10 concepts were curated against the 25,047 PT strings actually present rather than against a current MedDRA release, and split into two tiers.

Every primary result uses the `core` tier: 10 PTs in 3 concepts — muscle destruction specific enough to be hard to report for any other reason (rhabdomyolysis, myoglobin release, muscle necrosis). The remaining 13 PTs in 7 concepts form the `broad` tier, which adds general myotoxicity and the creatine-kinase markers; it is far more powerful and far more confounded — MYALGIA alone carries 163,419 reports and is reported against almost everything — and is used for sensitivity analysis only. The two tiers are not interchangeable: `core` admits **42,058** event cases before the polypharmacy exclusion and `broad` admits **339,063**, an eight-fold difference, so a reader applying the wrong tier will not reproduce any count in this paper.

Two renamings occur within the concept area, both clean instantaneous switches: `BLOOD CREATINE PHOSPHOKINASE INCREASED` → `CREATINE KINASE INCREASED` at 2026Q2 (part of a vocabulary-wide event in which 1,907 PT strings make their last appearance in 2026Q1), and `IMMUNE-MEDIATED NECROTISING MYOPATHY` → `IMMUNE-MEDIATED MYOSITIS` at 2019Q4. The latter is *not* meaning-preserving — the successor carries ~5× the per-quarter volume — so both sit in the broad tier.

3.6 The positive control set, and its verification

16 pairs, assembled from FDA product labelling: seven simvastatin pairs, two lovastatin, two atorvastatin, two rosuvastatin and three colchicine, against eight perpetrator drugs.

Every pair was checked against label text rather than asserted. An earlier version of this work carried `citation_status: to_verify` on all sixteen rows of the control file — a field created with the intention of checking them, never filled in — while §2 interrogated the *evaluation* reference at length. The check (`faers_ddi.verify_controls`, run against the same cached label corpus as §3.7) asks whether either drug's label names the other, whether a myotoxicity term appears within 600 characters of that mention, and whether the mention sits in contraindication or dose-limiting language:

status	pairs
named, myotoxicity-relevant, **and contraindicated or dose-limited**	**14**
named and myotoxicity-relevant	2
named only, or not found	0

All 16 are confirmed. Two — colchicine + cyclosporine and colchicine + atorvastatin — carry a myotoxicity warning without explicit dose restriction, and colchicine + atorvastatin is additionally the one pair sourced from case reports rather than labelling and graded `probable` rather than `established`. It is retained and flagged rather than dropped: removing a control because it scored poorly would be selection on the evaluation set.

The 16 are not 16 independent trials. They are five victim drugs, and simvastatin appears in seven of them. Recovery intervals are therefore computed by resampling the victim drug (§4.3), not by a binomial on the pair count.

3.7 Independent interaction reference

Because the authors wrote both the positive control set and the list defining "known pair", a second annotation was built that they did not author. Signals are evaluated against **three annotations** of increasing independence — the authors' own curated list, FDA labelling for any endpoint, and FDA labelling restricted to this endpoint — each applied both to all pairs and to pairs containing no positive-control drug. For each of the 200 screened ingredients we retrieved the most recent FDA product label via openFDA and recorded which other screened ingredients are named in its DRUG INTERACTIONS, CONTRAINDICATIONS or WARNINGS AND PRECAUTIONS sections. A pair is `label_documented` when either drug's label names the other. Labels are cached so the reference is fixed against future label revisions.

This is independent of the authors but not of FAERS. Labelling is informed by post-marketing surveillance, so it cannot establish that a signal was found independently of the data; it establishes only that the annotation was not written by us, which is the circularity at issue. Labels also warn by class ("strong CYP3A4 inhibitors") as often as by name, so the reference is under-sensitive, biasing measured enrichment downward.

That is the conservative direction for a claim that enrichment exists, and the anti-conservative direction for this paper's actual claim, which is that it does not. Attenuating a ratio toward unity moves it toward our own conclusion, so under-sensitivity cannot be offered as a safeguard here. §4.5 therefore reports the analysis restricted to pairs whose two labels both exist, which removes the part of the insensitivity that is structural rather than editorial.

3.8 Statistical measures

The eight cells of the 2×2×2 table are recoverable from the triple count, six marginals and the total.

Write n_{111} for the count of cases reporting drug A, drug B and event Z; $n_{11\cdot}$ for the co-report count; $p_A = P(Z \mid A)$, $p_B = P(Z \mid B)$ and $p = P(Z)$ for the two marginal risks and the database background; and α for the shrinkage constant. Both statistics compare n_{111} against an expectation and differ **only** in how that expectation is formed.

Ω (multiplicative null). The expectation E_{111}^{mult} is the fit of the log-linear model $[AB][AZ][BZ]$ — all pairwise associations, no three-way term:

$$\Omega = \log_2\left(\frac{n_{111}+\alpha}{E_{111}^{\text{mult}}+\alpha}\right)$$

fitted exactly by iterative proportional fitting rather than the published closed-form approximation, which we measured at up to 237% error once pairwise log-odds reach 2 — enough to produce $\Omega < -1$ on tables with no synergy.

Ω_add (additive null), primary. The expectation is the joint risk under additivity of excess risk, floored at the larger single-drug risk and capped at 1, applied to the co-report count:

$$E_{111}^{\text{add}} = n_{11\cdot}\,\min\bigl(\max(p_A + p_B - p,\ p_A,\ p_B),\ 1\bigr)$$

$$\Omega_{\text{add}} = \log_2\left(\frac{n_{111}+\alpha}{E_{111}^{\text{add}}+\alpha}\right)$$

The floor is not at zero: when both drugs are reported with the event *less* often than background, $p_A + p_B - p < 0$, and clipping to zero would make the expected count zero and Ω_add unbounded in the observed count. The cap at 1 binds precisely in the drug-dominant regime this paper is about, where $p_A + p_B$ can exceed unity. Shrinkage is identical for both, so the two estimands differ only in the null.

Ω_{025} and $\Omega_{\text{add},025}$ denote the 2.5th percentile of the corresponding gamma-Poisson posterior.

Shrinkage constant. $\alpha = 0.5$ is conventional and could not be verified against Norén et al. It is therefore varied across a 20-fold range as a sensitivity analysis (§4.8) rather than assumed.

Intervals and tests. All proportions carry Jeffreys 95% intervals; ratios carry log-scale intervals. Binomial tests are **not** used for inference: with 200 drugs each drug sits in 199 pairs, so pair outcomes are strongly dependent. Significance is assessed by a permutation test that holds the pair graph and the observed signal pattern fixed and randomises which drugs are annotated as implicated (10,000 permutations).

What "nominal 2.5%" does and does not mean. Ω_{025} is the 2.5th percentile of a gamma-Poisson posterior rather than a frequentist test statistic, so "nominal 2.5%" is a convention of the field and not a guarantee of the estimator; a rate departing from it is evidence about behaviour in this regime, not a violated guarantee. We use the convention because it is the operating point practitioners use, and because the comparisons below hold the cut fixed across both measures.

Design. Unit of analysis is the case; denominator is the deduplicated set less the polypharmacy exclusion, retaining cases with no resolved drug as background; drug roles are primary/secondary suspect plus interacting.

4. Results

4.1 Ω fails; both nulls mispredict in the same direction (Figures 1–2)

Ω recovered **4/16** positive controls at $\Omega_{025} > 0$, against **12/16** for the additive null at the same threshold. Pooled over the generated negative controls the two nulls' false-positive rates are 6.4% and 6.7%. **Those pooled rates do not describe the regime in which the recovery comparison is made**, and an earlier version of this manuscript presented them as though they did, calling the comparison matched. §4.3 shows they do not: among pairs as strongly associated as the positive controls the rates are 2.2% and 9.3%, and once they are equalised the recovery gap falls from eight pairs to one or two.

Simvastatin + amiodarone, named in advance as the pair that must work, scored $\Omega = \mathbf{-0.385}$ ($\Omega_{025} = -0.630$): 145 events among 649 co-reports against 189.5 expected under the multiplicative null.

The drugs of interest are the dominant reported causes of the outcome, with marginal relative risks of 3–19 against a 0.207% background, so the multiplicative null predicts very high joint rates (Figure 1):

pair	observed	multiplicative	additive
gemfibrozil + simvastatin	56.2%	73.9%	28.3%
atorvastatin + gemfibrozil	14.9%	72.6%	23.1%
amiodarone + simvastatin	22.3%	29.2%	11.0%

56% of gemfibrozil + simvastatin co-reports carry rhabdomyolysis and the pair still scores as protective.

The marginal-strength gradient is real, but it is not diagnostic of the multiplicative null. Ω correlates with $\log_2(\mathrm{RR\_A} \times \mathrm{RR\_B})$ at $\mathbf{r = -0.63}$ **(n = 16, 95% CI −0.86 to −0.19, p = 0.009)** (Figure 2). Because $\Omega = \log_2((\mathrm{O}+\alpha)/(\mathrm{E}+\alpha))$ and E is an increasing function of the same marginals that form the x-axis, part of any such correlation is induced by the construction — regressing a ratio on a proxy for its own denominator. We measured how much: drawing the triple count from each null's own expectation and recomputing the statistic 10,000 times gives an induced correlation centred on zero (median +0.03, 95% interval −0.23 to +0.26 for Ω). The observed −0.63 lies outside that interval, so the gradient is **not** an artifact of the estimator.

It is, however, **not specific to the multiplicative null.** The additive null — this paper's remedy — shows the same gradient, slightly stronger:

quantity regressed on $\log_2$(RR_A × RR_B)	r (95% CI)	p
observed event rate among co-reports	**+0.12 (−0.40 to +0.58)**	**0.67**
expected rate, multiplicative null	+0.94	—
expected rate, additive null	+0.94	—
Ω (multiplicative)	−0.63 (−0.86 to −0.19)	0.009
Ω_add (additive)	**−0.65 (−0.86 to −0.22)**	**0.007**

The observed joint event rate rises far more shallowly with marginal strength than either null predicts. The observed gradient is not distinguishable from zero, but its interval (−0.40 to +0.58) admits a moderate rise, so the claim is that the gradient is much shallower than predicted rather than that it is absent. Both nulls predict a steep rise; both are therefore wrong in the same direction, and the gradient in Ω reflects a property of the data — joint risk saturates as the individual risks grow — rather than a defect peculiar to multiplicativity. It will appear in any statistic that divides an observed rate by a marginal-driven expectation.

> An earlier version of this manuscript reported this correlation for Ω alone and read it as evidence that the multiplicative null is uniquely broken. That reading was incorrect. The correlation survives the artifact check, but it does not separate the two nulls, and it is reported here for both.

What *does* separate them is **level, not slope**: the multiplicative expectation is far larger at every point (72.9% vs 27.9% for gemfibrozil + simvastatin), so the same saturation drives Ω below zero while leaving Ω_add above it. Departure from additivity is the standard criterion for clinically meaningful interaction (Rothman, 1976; VanderWeele and Knol, 2014); departure from multiplicativity is a stricter and different question, and on a drug-dominant event it is a question almost nothing passes.

4.2 When the two nulls stop being different tests

The comparison in §4.1 has a structural explanation, and it is checkable before any outcome data is used.

This is a refinement of a known relationship, not a new one. Jung and Jung (2024) state that satisfying the multiplicative assumption implies satisfying the additive one, and draw the threshold consequence — that the additive model signals earlier and the multiplicative model marks stronger interactions. They state it without conditions. It holds under the positive single-drug effects their simulation assumes, and fails otherwise; what follows gives the condition, shows that real screens violate the unconditional form, and quantifies where it binds.

Both statistics are $\log_2((n + \alpha)/(E + \alpha))$ under identical shrinkage (§3.8), so they differ **only** through the expectation. The gamma-Poisson posterior quantile is decreasing in E, which gives an exact implication:

$$E_{111}^{\text{mult}} \;\geq\; E_{111}^{\text{add}} \quad\Longrightarrow\quad \Omega_{025} \leq \Omega_{\text{add},025} \quad\Longrightarrow\quad \{\Omega_{025} > 0\} \subseteq \{\Omega_{\text{add},025} > 0\}$$

When are they ordered? Writing a = RR_A and b = RR_B, the multiplicative expectation is proportional to ab and the additive one to $a + b - 1$, so

$$E_{111}^{\text{mult}} - E_{111}^{\text{add}} \;\propto\; (a-1)(b-1)$$

and the ordering holds exactly when both marginals fall on the same side of unity. Both drugs elevated is therefore sufficient, and the data agree without exception: among the 16,138 generated negative controls the unconditional form fails for **163** pairs and for **none** of the **6,163** where both marginals exceed 1. (The remaining failures are pairs where both drugs are *protective*; there the floor at max(p_A, p_B) in E_add binds and the proportionality above no longer governs.)

Where the expectations are ordered that way, the multiplicative null cannot signal on a pair the additive null misses. The two are then **not competing tests but one ordering read at two thresholds**, and any recovery comparison between them measures the threshold, not the null.

Whether the antecedent holds is empirical, and it depends on precisely the quantity this paper is about:

pairs	n	E_mult ≥ E_add
expected joint rate > 5% (drug-dominant)	4,314	**98.3%**
expected joint rate 0.5% (diffuse)	3,214	**16.9%**
all screened pairs	17,375	76.0%

The implication holds without exception in the data: of the **13,208** pairs whose expectations are ordered, **not one** signals under Ω while failing under Ω_add. All **64** pairs that do so — anywhere in the screen — have the expectations reversed, which is the only way the theory permits. Across the 17,375 screened pairs and the 19,826-pair high-marginal pool, there is no counterexample.

This is the diagnostic. Both expectations are functions of the marginals and the co-report count alone — no outcome comparison, no control set, no calibration. A screen can therefore determine *before looking at any result* whether its two candidate nulls are genuinely different tests or the same ordering at two operating points. When they are the same ordering, the only decision left is the threshold, and §4.3 shows that decision cannot be made from a nominal credibility bound.

It also explains §4.1 without appeal to any property of rhabdomyolysis: the multiplicative expectation exceeds the additive one exactly when both marginals are strong, which is the definition of the regime where DDI methods are validated.

4.2 The estimand switch does not inflate sensitivity

The additive null was adopted because it recovered more controls, and control recovery is then reported as validation — selection on the evaluation set. The selection decision is binary and was therefore cross-validated: the additive null wins in **16/16 leave-one-out folds**, and held-out recovery equals in-sample recovery (**12/16, optimism 0.000**).

The optimism figure carries less information than it appears to. When the same null wins in every fold, the held-out score for each control is by definition its in-sample score, so optimism is *identically* zero — it cannot take another value. It is a consequence of the selection being stable, not independent evidence for it, and an earlier version of this manuscript presented it in the abstract as though it were a measurement. What leave-one-out does establish is the stability itself: the choice between the two nulls does not depend on any single control.

It does **not** address the deeper issue that these 16 controls were chosen by the authors. That is addressed instead in §4.10, with a control set drawn by FDA labelling rather than by us, and it is where the sensitivity estimate actually degrades. See also §6.

4.3 Tier A and Tier B

Under the additive null, **12/16 controls signal (12/14 powered)**, against **4/16 for Ω (4/14 powered)** at the same threshold. Six of seven simvastatin pairs and three of three colchicine pairs recover; both lovastatin pairs have n_ab of 13 and 1 and are unmeasurable. Both denominators are given because they are not interchangeable: an earlier version quoted the additive count out of 14 and the multiplicative count out of 16 in the same comparison.

The interval must respect the control set's clustering. The 16 controls are five victim drugs and simvastatin appears in seven of them (§3.6), so they are not 16 independent trials. Resampling the victim drug rather than the pair:

estimate	naive binomial	**cluster bootstrap (victim drug)**
12/14 powered = 86%	62–97%	**50–100%**
12/16 all = 75%	51–91%	**30–96%**

The naive interval is roughly 40% too narrow. We report the clustered one. This is the same correction already applied to the screen enrichment, where pairs share drugs; applying it there and not here was two standards for one dependence structure.

The full specification grid. Two binary design choices were available — the event tier (`core`, specific; `broad`, inclusive) and the drug role policy (`primary`, suspect roles only; `sensitivity`, wider). `core/primary` is pre-specified in `config.yaml` as `event.primary_tier` and is the arm reported throughout. All four were computed, and an earlier version of this manuscript reported the pre-specified arm without disclosing that the others existed:

tier	role policy	additive	multiplicative	additive advantage
core	**primary** *(pre-specified)*	**12/16**	**4/16**	**+8**
core	sensitivity	12/16	4/16	+8
broad	primary	11/16	4/16	+7
broad	**sensitivity**	**6/16**	**6/16**	**0**

Under the widest event definition combined with the widest role policy, the additive null's advantage disappears entirely. The contrast holds in three of four arms and is null in the fourth. Any reader weighing the headline result should weigh that: it is robust to either widening alone, and not to both together. The broad tier adds 13 PTs across 7 concepts — the creatine-kinase markers, the two non-meaning-preserving renamings, and general myotoxicity terms including MYALGIA, which carries 163,419 reports on its own and is reported against almost everything (§3.5) — while the sensitivity policy admits concomitant drugs with no suspected causal role. The fourth arm therefore combines the least specific outcome available with the least specific exposure, which is the most likely reason both nulls converge there rather than one degrading faster than the other. It is the least specific analysis available — but it is not unreasonable, and it is now on the record.

Against the **full pool of 16,138 negative controls** (not a sample — sampling 2,000 left the threshold at the mercy of the draw, moving it by more than a factor of four across legitimate runs), the pooled false-positive rates at $\Omega_{025} > 0$ are **6.7%** additive and **6.4%** multiplicative, against a nominal 2.5%. Reported by stratum, and for both nulls:

stratum	n	additive	multiplicative	ratio
easy (neither drug RR $\geq$ 2)	6,674	4.02%	4.94%	0.81
hard (at least one RR $\geq$ 2)	9,464	8.54%	7.50%	1.14
all	16,138	**6.67%**	**6.44%**	1.03

An earlier version of this manuscript reported the strata for the additive null alone and described the near-identical pooled rates as making the §4.1 recovery comparison “essentially matched”. **Both statements are withdrawn.** The pooled figures are the average of two differences running in opposite directions, and — more importantly — they are measured almost entirely outside the regime in which recovery is measured.

The negative and positive control populations barely overlap. The generator excludes any pair in which *both* drugs are on the implicated list, a reasonable guard against seeding the null set with true positives, but every positive control is exactly such a pair — so the generator **cannot produce a negative that resembles a positive**, and that is a property of the exclusion rule rather than of the data. Positive controls sit at a median $\log_2$(RR_A × RR_B) of **8.23** (IQR 7.92–8.99); generated negatives at **0.56** (IQR −1.52–2.44), and **only 11 of 16,078 (0.1%) reach the positive controls’ interquartile floor.** Since §4.1 establishes that the expected count rises steeply with marginal strength, a pooled rate is not the rate that applies to the pairs being recovered. Across quintiles of marginal strength the additive rate runs 0.93%, 3.55%, 7.21%, 10.86%, 10.91% and the multiplicative 1.62%, 5.72%, 8.64%, 10.39%, 5.97%.

A purpose-built pool for the regime under study. All pairs among the 1,577 ingredients with RR $\geq$ 2 and at least 20 co-reports, excluding the positive controls, every pair documented in the endpoint-specific label reference, and both-implicated undocumented pairs as too likely to be unrecorded true interactions — 19,826 pairs. Across the whole pool the additive null fires on **7.2%** and the multiplicative on **3.7%**. Restricted to the **2,345** pairs at positive-control strength the rates are **9.3%** and **2.2%**, against 9.0% and 1.2% on the 166 in-regime pairs the standard generator happens to yield.

Those 2,345 pairs are drawn from 478 drugs and each drug recurs across many of them, so the interval must respect that dependence as it does elsewhere in this paper. Resampling drugs rather than pairs (20,000 draws):

null	rate	cluster bootstrap 95% CI	naive binomial (too narrow)
multiplicative	52/2,345 = **2.2%**	**1.24–3.34%**	1.68–2.87%
additive	219/2,345 = **9.3%**	**7.29–11.32%**	8.21–10.57%

Against a nominal 2.5%, Ω runs at close to its advertised rate in this regime (2.2%, interval covering 2.5%) while Ω_add runs close to fourfold above it (9.3%, interval

excluding it). The clustered interval is nearly twice the width of the binomial one, and the conclusion does not depend on which is used for the additive null — but for the multiplicative null it is the difference between an interval that comfortably covers the nominal rate and one that barely does. The miscalibration is one-sided: on error rate alone, only the additive null is out of specification. What disqualifies Ω here is power rather than size — it is systematically negative in this regime, so it holds its nominal rate by almost never firing.

Recovery at matched in-regime error rates. Calibrating each null against the in-regime negatives separates the choice of null from the choice of operating point:

operating point	additive	multiplicative	gap
$\Omega_{025} > 0$ (as published)	12/16 @ 9.0%	4/16 @ 1.2%	**8**
matched at 5% in-regime FPR	12/16	10/16	**2**
matched at 10%	12/16	11/16	**1**
matched at 20%	14/16	12/16	**2**

The eight-pair gap becomes one to two once error rates are equalised. The additive null still wins at every matched rate, so the direction is real, but most of the apparent advantage comes from $\Omega_{025} > 0$ being a far stricter operating point for Ω than for Ω_add in this regime: the multiplicative null must move to about -1.4 to reach 9% in-regime, precisely because Ω is systematically negative here (§4.1). With 16 controls in five victim-drug clusters and a 50–100% interval on the unmatched estimate, a one-to-two-pair difference is not separable from noise and we do not claim it is.

Caveat. The matched-rate calibration rests on the 166 in-regime pairs of the generated pool, so the 5%/10%/20% points sit on roughly 8/17/33 of them. The purpose-built pool confirms the rates at 2,345 pairs but the matched-recovery table has not been recomputed on it, and the in-regime cut — the marginal strength of the weakest positive control — is our choice, not pre-specified.

The calibrated threshold is validated out of sample. A quantile of the pool whose false-positive rate is then reported returns the target by construction, so the 5% is definitional rather than measured. Splitting the pool 500 times — calibrating on one half, measuring on the other — gives a held-out threshold of **+0.429** and a held-out false-positive rate of **5.03% (95% CI 4.37–5.74%)** against the in-sample **+0.436**. The in-sample calibration is therefore very nearly unbiased, but the rate is now a measurement. Downstream, the pairs expected by chance among the 17,375 screened become **874 (95% CI 759–997)** rather than the 869 implied by the nominal 5%.

Sensitivity is quoted at $\Omega_{025} > 0$ throughout, not at the calibrated threshold. At +0.436 recovery is 11/15 of the controls that enter the screen (§4.5). The two operating points are reported separately and should not be combined.

Fifteen of the sixteen controls enter the screen. Itraconazole + lovastatin does not. Both ingredients are among the 200, but the pair is co-reported once in 22 years and so falls below the three-co-report floor fixed in advance. Recovery is therefore quoted out of 16 where the control set is the unit and out of 15 where the screen is, the two denominators are not interchangeable, and the missing pair is the least powered of the sixteen rather than a failure to detect.

4.4 Polypharmacy leverage (Figure 5)

The strongest apparent false positive, measured before the cap was applied, was alirocumab + ipratropium: 88 co-reports, 88 events. The pair does not appear in the shipped screen or negative-control tables, both of which are generated after the exclusion; it is quoted here from the uncapped run that motivated the cap. Those 88 cases share 5 distinct event dates and 1 distinct age, each listing 31–40 drugs — residual near-duplicates the exact-set fingerprint could not merge.

A case listing 40 drugs contributes 780 pairs. **19,005 cases (0.09% of the database) contribute 34.7% of all drug pairs at a 4.3× enriched event rate.** That aggregate hides a reversal, and Figure 5 shows it: the event rate climbs across the bands to **1.44% at 31–50 drugs**, then collapses to **0.03% at 51+ drugs** — roughly seven-fold *below* the 0.207% background — even though the 51+ band is the single largest contributor of pairs in the database at 18% of them. Leverage and enrichment are therefore not the same problem. The pair arithmetic is what motivates the cap; the enrichment applies to the 21–50 bands and not to the largest one, whose reports look more like registry dumps than clinical narratives. Capping at 20 drugs per case improved sensitivity (11→12/16) and the false-positive rate (6.9%→6.7%) simultaneously, and the multiplicative null improves from 2/16 to 4/16, so the headline contrast in §4.1 is *understated* by the cap rather than produced by it. Nothing in the positive controls could have revealed the leverage problem itself — it is visible only from the false-positive side. **The cap value, however, was chosen by looking at control recovery**, and §4.9 reports the full sweep: the conclusion is flat from 15 to 40, and a cap of 10 would have been better on both axes. We retain 20 rather than the dominating value: re-tuning the cap on the same 16 controls used to measure performance would convert a pre-specified parameter into a fitted one, so the reported configuration is deliberately not the best available on our own numbers.

4.5 The screen shows no demonstrable enrichment for genuine interactions (Figures 3–4)

Not every entry in the vocabulary can form a drug pair. The selection rule is applied mechanically to FDA's resolved active-ingredient field, and four of the 200 entries are placeholders that field supplies where it cannot resolve a moiety — `UNSPECIFIED INGREDIENT`, `HERBALS`, `INSULIN NOS` and `CANNABIS SATIVA SUBSP INDICA TOP`. A further three pairs are one moiety with itself, valproate reaching the vocabulary as `VALPROATE`, `VALPROIC ACID` and `DIVALPROEX`. Together these account for **719 pairs (4.1% of the screen), of which 38 signal and 122 fall in the `plausible` band**. They are not removed from the primary analysis — the drug-selection rule was fixed in advance, and excluding terms after seeing results would be a researcher degree of freedom — and the sensitivity is small: `plausible` enrichment moves from **0.766 to 0.752** and `known_pair` from **2.015 to 1.995** when they are dropped. The band conclusions do not depend on them. They do matter for reading individual pairs, since a pair naming `UNSPECIFIED INGREDIENT` is uninterpretable however it scores.

17,375 pairs tested, **1,022** above threshold. A pooled chance expectation — the held-out false-positive rate of 5.03% applied uniformly — gives **874 (95% CI 759–997)**, suggesting a modest excess.

That calculation assumes the false-positive rate is constant across the screen, and §4.3 shows it varies by an order of magnitude. The four bands differ systematically in marginal strength (median $\log_2$(RR_A × RR_B) of 2.85, 4.16, 5.32 and 8.15). Matching each screened pair to the observed rate among negative controls of similar marginal strength:

band	tested	observed	pooled (5.03%)	strength-matched
unsupported	11,887	717	598	**824**
plausible	4,930	228	248	**348**
known pair	543	66	27	**39**
positive control	15	11	1	1
total	17,375	**1,022**	**872**	**1,212**

Under the strength-matched expectation the screen returns fewer signals than chance predicts — 1,022 against 1,212 — and only `known_pair` exceeds its own expectation. The apparent excess under the pooled figure is an artefact of averaging a rate that varies widely. This sharpens the negative result below rather than threatening it.

But the 1,212 figure is far weaker than the 1,022 it is compared against, and we state how weak. The negative pool's top quintile spans marginal strength 2.80 to 8.95, and the median pair of *every* band in the screen — 2.85 unsupported, 4.16 plausible, 5.32 known pair, 8.15 positive control — falls inside it. More than half the screen therefore collapses into a single bin and inherits a single rate (7.06%), which is why 17,375 × 7.06% reproduces the strength-matched total to within about one percent. The column is closer to one extrapolated number applied to the whole screen than to a genuine per-pair matching, and that number is estimated from negatives concentrated near the bottom of the bin. The rate is also non-monotone at the top — 0.78%, 2.83%, 5.72%, 8.71%, 7.06% across quintiles — so it is not even clear that extrapolating upward is conservative. We report the comparison because the uniform assumption is demonstrably wrong and the direction of that error is not in doubt; nothing rests on the precise size of the shortfall.

Multiplicity. Ω_add,025 is a shrinkage bound, not a p-value, and carries no family-wise or false-discovery guarantee; the calibrated threshold stands in for one. As an independent check, a one-sided Poisson test of each triple count against its additive expectation, with Benjamini–Hochberg control at q = 0.05, yields **1,147 discoveries** — and every one of the 1,022 shrinkage signals is among them. The shrinkage threshold is the more conservative of the two rules, so no conclusion below depends on the absence of a formal correction.

band	signalled	rate	enrichment (95% CI)
positive control	11/15	73.3%	12.16 (8.89–16.63)
known pair	66/543	12.2%	2.02 (1.59–2.55)
plausible	228/4,930	4.6%	**0.77 (0.66–0.89)**
unsupported	717/11,887	6.0%	1.00 (0.90–1.11)

A drug-level permutation test gives pooled enrichment 2.29× (p = 0.0012) under the **author-curated** annotation, so that pooled effect is not an artefact of pair dependence alone. **Under the independent FDA-labelling annotation the same test is not significant** (2.80×, p = 0.14). The distinction matters and an earlier version reported only the first: dependence-aware significance is present for the annotation the authors wrote and absent for the one they did not.

But the annotation is not independent of the control set. All 12 positive-control drugs are among the 64 drugs defining "known pair", and the list was written by the same authors. Restricting to pairs containing **no** positive-control drug:

	signalled	enrichment (95% CI)
known pair, no control drug	16/237	**1.12 (0.69–1.81)**
unsupported	717/11,887	1.00

Enrichment is indistinguishable from unity. The pooled 2.02× is essentially entirely attributable to pairs containing a drug the authors had already nominated.

Confirmed against an independent reference. To remove any dependence on the authors' curation we built a second annotation from FDA product labelling (§3.7): 6,106 pairs in which one drug's label names the other. It captures 16/16 positive controls, so it is not insensitive.

Two corrections were required before it could be used. First, **a label documents that two drugs interact, not that the interaction causes this event**: 82% of the name-matched pairs are documented for an unrelated endpoint, and omeprazole + warfarin — a real CYP2C19 interaction affecting INR — was being counted as a hit in a myotoxicity screen. The endpoint-specific reference additionally requires a myotoxicity term within 600 characters of the partner drug's name, giving 709 pairs which still capture 16/16 positive controls. Second, **documented pairs are co-reported about three times more often than undocumented ones** (median n_ab 202 vs 69, Mann-Whitney $p = 2{\times}10^{-75}$), and co-report count drives statistical power directly, so the crude comparison confounds "documented" with "well powered". Results are therefore also reported stratified on co-report count decile (Mantel-Haenszel).

annotation	scope	signalled	enrichment (95% CI)	stratified
author-curated	pooled	66/543	2.02 (1.59–2.55)	—
author-curated	no control drug	16/237	**1.12 (0.69–1.81)**	—
FDA labelling, any endpoint	pooled	110/1,339	1.44 (1.19–1.75)	1.24
FDA labelling, any endpoint	no control drug	57/1,069	0.92 (0.71–1.2)	0.75
FDA labelling, myotoxicity	pooled	48/240	3.52 (2.71–4.56)	3.08
FDA labelling, myotoxicity	**no control drug**	10/142	**1.23 (0.67–2.24)**	**1.045**

The reference is structurally blind to part of the screen. 11 of the 200 screened ingredients (**5.5%**) have no FDA label in openFDA at all, so no pair containing one can ever be `label_documented`. Because those 11 are well co-reported they touch a disproportionate share of the pair space: **1,712 of 17,375 pairs (9.9%)** are undocumentable by construction and fall into the denominator. The gap is not random — it falls on agents without a current US marketing authorisation, and the one that matters for this endpoint is **fusidic acid**, whose statin combination is contraindicated in practice and which is the screen's highest-ranked pair by event rate (§4.7).

Widening the reference beyond the screened set makes the coverage problem more visible without changing the screen: of the 800 ingredients for which labels were retrieved for the screen-size sensitivity analysis (§4.9), 138 (17.2%) have none, and those include **cerivastatin** — a statin withdrawn worldwide for rhabdomyolysis with gemfibrozil — together with **bezafibrate, ciprofibrate** and **telithromycin**. None of the four entered the top-200 screen, so they do not bear on the result reported here; they indicate that any wider screen would inherit a larger version of the same blindness.

Restricting to non-control pairs where *both* labels exist, so that "undocumented" means the label is silent rather than absent, gives enrichment **1.24 (0.68–2.26)** crude and **1.067 (0.292–1.972)** stratified on 142 documented pairs. **The negative result is unchanged by the correction**, but the reference's coverage is a property of US marketing status rather than of pharmacology and should be read that way.

Note the direction of this bias. §3.7 states that under-sensitivity biases enrichment downward, "the conservative direction for a claim that enrichment exists" — which is true of the pooled 3.52×. It is the **anti**-conservative direction for this paper's actual claim, which is that enrichment is absent: misclassifying documented pairs as undocumented attenuates the ratio toward unity, and unity is the conclusion. That is why the coverage-restricted analysis above is reported rather than the caveat alone.

Power. This comparison rests on 142 documented non-control pairs. At the observed baseline signal rate it has 83% power to detect a true enrichment of 2.5× but only 57% at 2.0× and 23% at 1.5×. The correct reading is therefore **"no enrichment above roughly 2.24×"** — the upper confidence bound — rather than "no enrichment". A real but modest enrichment would very likely be missed here, and excluding it would need either a larger reference or a screen extended beyond the top 200 drugs.

A note on precision. The crude `plausible` enrichment appears in this paper as 0.77, 0.767 and 0.766. It is one quantity — 228 signals in 4,930 pairs against the `unsupported` rate — computed in three places that round differently: the band table reports two decimals, the marginal-strength stratification three, and the vocabulary-hygiene sensitivity three from its own recomputation over the same pairs. The spread is rounding, not disagreement, and no conclusion turns on the third decimal.

All annotations agree. Pooled enrichment is real and, under the endpoint-specific reference, substantial (3.52×). **It vanishes once pairs containing a positive-control drug are removed** — 1.23× crude, **1.045 after adjusting for co-report count**. The negative result does not depend on who wrote the reference, on whether the reference is endpoint-specific, or on the power confound.

The `plausible` band — one implicated drug plus an unimplicated partner, designated in advance as where a novel interaction would appear — has enrichment **0.77 (0.66–0.89)**, significantly *below* unity.

Adjusted for marginal strength, which the bands differ on. §4.1 shows the expected count rises steeply with $\log_2$(RR_A × RR_B), and the bands are not exchangeable on it: median 2.85 (`unsupported`), 4.16 (`plausible`), 5.32 (`known_pair`), 8.15 (positive controls). Comparing raw signal rates across a 1.3-unit gap in a covariate known to drive the statistic needs adjustment, and an earlier version stratified only on co-report count. Applying the same Mantel–Haenszel machinery to deciles of marginal strength, with the drug-level cluster bootstrap:

band	crude	stratified on marginal strength (95% CI)
`plausible`	0.767	**0.749 (0.567–0.973)** — still below unity
`known_pair`	2.015	1.835 (0.99–2.989) — **now includes unity**

The confound is real but small, and it does not run the way one might guess: signal rate by marginal-strength quintile is weak and non-monotone (3.7%, 7.2%, 6.6%, 5.2%, 6.7%). **The `plausible` band's deficit survives adjustment.** The `known_pair` band's apparent 2× does not survive it intact — the dependence- respecting interval now touches unity — which is consistent with the circularity argument above rather than an additional finding.

4.6 Temporal stability: composition, not count (Figure 4)

Requiring the signal in all three eras reduces 1,022 pairs to **19**.

What this filter is. Each era bin is scored against the *same* threshold (+0.436), calibrated on the full 22 years. A bin holds roughly a third of the cases, so the shrinkage bound is systematically lower in every bin and the filter is far stricter than "the signal is present in each era" — it is "the signal clears a full-data threshold three times on third-power data." It therefore selects on co-report count as much as on temporal consistency. This does not affect the count comparison below, because the identical filter is applied to the negative controls, but it does mean era-stability is not a pure measure of temporal persistence and should not be read as one.

eras with signal	pairs
3 of 3	19
2 of 3	170
1 of 3	778

Of the 19, nine carry prior support. But the same filter applied to the 6,471 negative controls that also enter the screen admits **6 of them (0.093%, 95% CI 0.039–0.191%)**, implying **16.1 era-stable pairs by chance (95% CI 6.7–33.2) against 19 observed**.

The number of era-stable pairs is not distinguishable from chance. An earlier version of this analysis reported this filter as the paper's principal contribution, on the basis of an enrichment figure computed without ever applying the filter to negative controls. That was wrong.

What appeared to survive was a composition-based claim: among the era-stable pairs, 10 of 1,339 label-documented pairs signal against 9 of 16,036 undocumented ones — enrichment **13.31× (95% CI 5.42–32.69)**, interval excluding unity. **That figure does not withstand the two corrections §4.5 applies to every other enrichment in this paper**, and an earlier version reported it without them: it uses the *any-endpoint* reference, shown two sections earlier to be 82% endpoint-irrelevant, and it is unstratified on co-report count.

Applying both:

reference	scope	documented signalled	crude (95% CI)	stratified (95% CI)
any endpoint	all pairs	10/1,339	13.31 (5.42–32.69)	—
any endpoint	no control drug	2/1,069	4.45 (0.90–22.0)	—
endpoint-specific	all pairs	8/240	51.9 (21.1–127.9)	32.6 (0.0–173.8)
endpoint-specific	**no control drug**	**0/142**	—	**0.0**

Once the reference is made endpoint-specific and control drugs are removed, not one era-stable pair is documented. The composition claim therefore rests entirely on pairs containing a drug the authors had already nominated — the same circularity as §4.5, which the earlier draft asserted this analysis had escaped. It had not. *How many* pairs survive the era filter is consistent with chance, and *which* pairs survive is not demonstrably non-random either.

4.7 Confounding explains the unsupported era-stable pairs, but not all of them

The 19 era-stable pairs comprise 5 positive controls, 4 `known_pair`, 2 `plausible` and 8 `unsupported`. Both of the last two groups are examined here; an earlier version of this manuscript examined only the 8.

The 8 unsupported pairs are statin proxies. Most carry a statin, fibrate or colchicine on **88–100%** of their event cases against a 40.5% background — paroxetine + valsartan, levothyroxine + valsartan, aspirin + metoprolol and aspirin + ramipril at exactly 100%. These are markers for “cardiovascular patient taking a statin”; the statin is the cause and the pair is a proxy.

Two adjustments were tried for inpatient confounding, and only the second is adequate.

A drug-list proxy, which is too small to be informative. Excluding cases containing any of 30 hand-picked procedural or critical-care agents (neuromuscular blockers, anaesthetics, vasopressors, IV fluids) **removes 275,205 cases — 1.4% of the analysis population** — and leaves every band enrichment essentially unchanged (`plausible` 0.76 → 0.75). An earlier version reported this as evidence that ICU confounding does not drive the result. **It is not evidence of that.** Perturbing 1.4% of the data cannot exclude a confounder, and a null was near-guaranteed before the analysis ran. (The earlier phrasing also read as though 275,205 were the *analysed* set; it is the excluded set.)

The reported outcome code, which is the right instrument. FAERS records hospitalisation directly: `outc_cod = 'HO'` covers 5,709,555 reports and was already parsed but unused. Stratifying the screen on it splits the analysis population into genuinely comparable halves rather than shaving 1.4% off one end, and is reported in §4.9.

What the drug-list exclusion did show is that removing one confounder reveals the next: the replacement top hits were naloxone + zopiclone, which carries an overdose or impaired-consciousness term on 64.9% of its event cases against a 13.2% background. Rhabdomyolysis has many non-interaction causes, each with its own drug signature.

The same test does not dispose of the two `plausible` pairs, which is the band designated in advance as where a novel interaction would appear:

pair	n_ab	events	rate	Ω_add,025	event cases carrying a *third* implicated drug
atorvastatin + fusidic acid	185	155	**83.8%**	1.02	**13/155 (8.4%)** vs 48.7% background
ciprofloxacin + simvastatin	377	152	40.3%	1.66	68/152 (44.7%) vs 48.7% background

Atorvastatin + fusidic acid is the **highest-event-rate pair in the entire screen** — above every one of the 16 positive controls, the best of which (cyclosporine + simvastatin) reaches 71.0% — is era-stable across all three eras, exceeds its additive expectation, and is *under*-represented for third-drug polypharmacy rather than over-represented. It is also a real and serious interaction: systemic fusidic acid with a statin is contraindicated.

It appears in the `plausible` band because **neither reference contains it**. Fusidic acid is not approved for systemic use in the United States, so openFDA returns no label for it at all and no fusidic acid pair can ever be `label_documented` (§4.5); it is also absent from the authors' own implicated-drug list. The screen ranked a genuine contraindicated interaction first and both evaluation references were blind to it.

Ranking the screen, and separating signal from proxy. The same third-drug test applies to every signalled pair, not only the era-stable ones. Ranking all signalled pairs with at least 50 co-reports by event rate:

pair	co-reports	event rate	Ω_{025}	signals under Ω too	event cases with a third implicated drug
Atorvastatin + Fusidic Acid	185	83.8%	-0.27	no	8.4%
Abiraterone + Rosuvastatin	283	82.7%	+1.82	yes	16.7%
Fusidic Acid + Simvastatin	142	79.6%	-0.52	no	19.5%
Influenza Virus Vaccine + Simvastatin	174	78.2%	+0.23	yes	64.7%
Paroxetine + Rocuronium	55	74.5%	+3.09	yes	100.0%

The proxy test separates the list cleanly. The bottom two entries carry a third implicated drug on **64.7%** and **100%** of their event cases: these are the "cardiovascular patient taking a statin" pattern already described, and the pair is not the cause. The top three sit at **8.4%, 16.7% and 19.5%** — they carry their own signal.

Those top three are also **absent from every reference available to us**, and two of them are the same contraindication: fusidic acid with atorvastatin and with simvastatin, ranked first and third in the entire screen.

Abiraterone + rosuvastatin is the one entry that clears both nulls. Its Ω_{025} of **+1.82** is remarkable in context: §4.3 shows the multiplicative null is systematically negative in this regime and fires on only 2.2% of strongly-associated non-interacting pairs, so a pair clearing it here is clearing the conservative test where that test almost never fires. We report this as a ranked observation and **make no mechanistic claim**: the cached FDA label for abiraterone documents

CYP2D6 and CYP2C8 inhibition and says nothing about transporters, statins or muscle, and rosuvastatin is not a CYP2C8 substrate. Confounding by indication cannot be excluded either — abiraterone treats metastatic prostate cancer, and statin co-prescription in that population is common — although that would raise co-reporting rather than the event rate *conditional* on co-reporting, which is what is elevated here. It is a candidate for follow-up in a data source with exposure denominators, not a finding.

Revised conclusion. No pair identified by this screen constitutes a *novel* pharmacokinetic interaction — atorvastatin + fusidic acid is documented, not novel. But the earlier claim that every era-stable pair traces to confounding was too strong: the top of the ranking is a genuine severe interaction that the references could not see. That is a statement about the references at least as much as about the method, and it is the concrete form of the reference-quality limitation quantified in §4.5 and §4.10.

4.8 Conclusions do not depend on the unverified shrinkage constant (Figure 6)

$\alpha = 0.5$ could not be verified against the source. Varying it over a 20-fold range and recalibrating the threshold on the full negative-control pool at each value:

α	calibrated threshold	positive controls recovered	pairs signalled
0.1	+0.635	10/15	998

Control recovery varies by two pairs and the signal count by 7% across the range. No conclusion in this paper turns on the value of α. The constant remains unverified against Norén et al., but is no longer a live dependency.

4.9 Sensitivity analyses

> **These analyses are post-hoc.** None was pre-planned. Each was added in response to a specific objection raised during internal review, and several changed the paper's conclusions. Counting all seven rounds, this manuscript reports the outcome of roughly fifty distinct analyses. No correction has been applied across them, and the intervals below are therefore nominal. The two claims the paper rests on (§4.1, §4.4) were specified before any result was seen; nothing in §4.9 was.
>
> The most recent round added the induced-correlation simulation (§4.1), the held-out threshold calibration (§4.3), the Benjamini–Hochberg comparison and the reference-coverage restriction (§4.5), the endpoint-specific era-stable analysis (§4.6), the polypharmacy cap sweep (below) and the matched-threshold held-out control comparison (§4.10). Four of those **contradicted** claims in the previous version, and each is flagged where it appears.

Screen size and the power of the negative result. The negative result at top-200 rested on 142 documented non-control pairs. Extending the label reference to 800 drugs and widening the screen raises this to 442.

screen	pairs	documented non-control	crude enrichment	stratified (cluster bootstrap)
top-200	17,375	142	1.23 (0.67–2.24)	1.045 (0.32–2.034)
top-400	53,229	321	1.77 (1.18–2.65)	1.311 (0.599–2.117)
top-800	131,888	444	2.05 (1.39–3.04)	1.265 (0.601–1.988)

Two things matter here. The crude interval excludes unity at top-400 and top-800; the **cluster bootstrap, which resamples drugs rather than pairs, does not** at any size. With each drug appearing in hundreds of pairs, the pairwise interval is anticonservative, and the dependence-respecting interval is the one to read. **The negative result survives the power increase.**

Selection on the outcome. Screened drugs are chosen by co-reporting with the event. Reselecting the same number by total report volume instead gives stratified enrichment 0.368 (0.0–1.839) against 1.045 (0.32–2.034) for the primary selection. Both include unity — but **this check is close to uninformative and should not be read as reassurance.** Volume-based selection leaves only 38 documented non-control pairs, of which **one** signals; the crude interval runs 0.14–6.49, which is consistent with almost any effect. The honest statement is that this analysis has too little power to detect whether outcome-based selection manufactures the result, not that it does not.

Era bin definitions. The three-bin split was fixed by hand. Varying it:

definition	bins	era-stable pairs
3 bins (primary)	3	19
2 bins	2	84
4 bins	4	6
5 bins	5	3

The count of era-stable pairs is almost entirely a function of how many bins are demanded, from 84 at two bins to 3 at five. This is a further reason not to treat that count as a finding (§4.6).

Ingredient resolution accuracy. Coverage was reported in §3.4; accuracy is bounded here. Across 32,655 verbatim drug names carrying FDA's own `prod_ai` annotation on at least 20 rows, **98.99%** of rows agree with the modal ingredient for that name, and **95.9%** of names are annotated unanimously. Level-2 backfill copies that annotation, so this bounds its reliability.

The polypharmacy cap was chosen on the evaluation set. 20 drugs per case was adopted in §4.4 because it improved control recovery *and* the false-positive rate — a decision made by looking at the controls, which was not previously disclosed as such. Varying it:

cap	analysis cases	additive	multiplicative	FPR at $\Omega_{025} > 0$
10	20,202,853	**13/16**	4/16	**6.0%**
15	20,259,682	12/16	4/16	6.0%
20 (adopted)	20,274,416	12/16	4/16	6.7%

cap	analysis cases	additive	multiplicative	FPR at $\Omega_{025} > 0$
30	20,284,277	12/16	2/16	7.8%
40	20,288,002	12/16	2/16	7.6%
none	20,293,421	11/16	2/16	6.9%

Two things follow. Capping is justified — every capped arm beats the uncapped one on recovery, and the multiplicative null degrades from 4/16 to 2/16 without a cap, so the headline contrast is *understated* at cap 20. But **20 is not the optimum**: a cap of 10 is better on both axes (13/16 at 6.0%). The adopted value is a round number that was not tuned further, and the conclusion is flat across 15–40. No result in this paper turns on the choice, but the choice was made after seeing the controls.

Inpatient status, using the reported outcome code. §4.7 explains why the 30-drug proxy (1.4% of cases) cannot exclude inpatient confounding. FAERS records the outcome directly, and stratifying the whole screen on `outc_cod = 'HO'` splits the population into two large, comparable halves:

stratum	cases	event cases	event rate	`plausible`	`known_pair`
hospitalised	4,274,465	28,610	0.669%	**0.639**	1.839
not hospitalised	15,999,951	13,279	0.083%	**0.825**	1.885

Hospitalisation is strongly associated with myotoxicity reporting — an **8-fold** difference in event rate — so it is a genuine confounder, unlike the 1.4% drug proxy which could not have detected one. That earlier attempt perturbed 1.4% of the population and left band enrichment essentially unmoved, but **perturbing 1.4% of the data cannot exclude a confounder**: a null was near-guaranteed before the analysis ran, and we do not present it as evidence. **Both strata reproduce the result**: the `plausible` band sits below unity in each, and `known_pair` enrichment is essentially identical across them. The negative discovery result is not an artefact of inpatient case mix.

Demographic strata. No subgroup analysis had been done despite the fields being available.

stratum	cases	event rate	crude enrichment (pairwise CI)	stratified (cluster bootstrap)
female	10,686,771	0.136%	2.13 (1.2–3.78)	1.723 (0.534–3.447)
male	6,998,836	0.327%	0.69 (0.26–1.82)	0.542 (0.0–1.459)

Myotoxicity is reported at 2.4× the rate in male as in female reports, consistent with the known epidemiology of rhabdomyolysis. **Both cluster-bootstrap intervals overlap unity**, and so does the crude male one. The crude female interval does *not* — 2.13 (1.20–3.78) — and it is the only non-control enrichment anywhere in this paper whose interval excludes unity. We report it rather than only its clustered counterpart, because it is a clean instance of the anticonservatism §3.8 describes: 15,471 pairs built from a few hundred drugs are not independent trials, and resampling drugs moves the estimate to 1.723 with an interval spanning unity. No subgroup claim is made on either scale, and these subgroups are underpowered.

4.10 Recovery on controls the authors did not choose

The 16 positive controls were author-selected, and leave-one-out (§4.2) bounds optimism only in the choice of *estimand*, not of controls. A second control set was therefore drawn by FDA labelling rather than by us: every label-documented myotoxicity pair with at least 50 co-reports that is **not** among the 16.

Both nulls are scored at the same threshold. An earlier version of this table compared the multiplicative null at $\Omega_{025} > 0$ against the additive null at the calibrated +0.436 — an asymmetry that ran *against* the additive null, so the finding survived, but the row was not a like-for-like comparison.

control set	n	threshold	multiplicative	additive
author-selected	14 powered	$\Omega_{025} > 0$	4/16	**12/14 (86%)**
label-selected	349	$\Omega_{025} > 0$	29 (8.3%)	**55 (15.8%)**
label-selected	349	calibrated +0.436	15 (4.3%)	**42 (12.0%)**

The direction replicates at both operating points — additive recovers 1.9× as many pairs at threshold 0 and 2.8× at the calibrated threshold — but the recovery RATE collapses from 86% to 12–16%.

The gap is not statistical power. Co-report counts are statistically indistinguishable between the two sets (median 420 vs 321, Mann-Whitney p = 0.45). Recovery in the label-selected set *falls* as co-reporting rises — 19%, 17%, 10%, 2%, 0% across quartiles and the top decile — the opposite of a power effect.

The mechanism is the event rate among co-reports:

control set	median event rate among co-reports	vs database baseline
author-selected	29.2%	141.4×
label-selected	0.72%	3.5×
label-selected, top decile by co-reporting	0.12%	**0.57**×

The most heavily co-reported label-documented pairs show **no elevation of the event rate at all** — below the database baseline. These are common co-prescriptions whose labels carry a *class* warning ("use with caution with CYP3A4 inhibitors") rather than a pair-specific one. There is nothing in the data for any method to detect.

So neither figure is the sensitivity. **86% is an upper bound contaminated by selection for famous, well-reported interactions; 12% is a lower bound contaminated by a reference containing pairs with no detectable signal.** Pinning the value down needs a severity-graded reference that distinguishes pair-specific interactions from class warnings — which is precisely what DrugBank or a curated DDI compendium would supply, and what this study lacked.

4.11 Does the failure generalise beyond this event?

The paper's claim is conditional — Ω fails *when the drugs under study are the leading reported causes.* One event demonstrates the phenomenon, not the condition. Two further events were analysed.

The torsade PT list is curated to the same standard as the primary event — repolarisation-specific terms only. An earlier version also counted `CARDIAC ARREST`, `VENTRICULAR TACHYCARDIA` and `VENTRICULAR FIBRILLATION`, non-specific terminal events with many non-QT causes, which tripled the event rate (0.66% against 0.207% for rhabdomyolysis) and made the replication a looser test than the analysis it was replicating. Both lists are reported.

event	event rate	median marginal RR	multiplicative	additive	Ω vs $\log_2$(RR_A×RR_B)	Ω_add vs same
rhabdomyolysis (primary)	0.207%	24.3	4/16	12/16	r = −0.63, p = 0.009	r = −0.65, p = 0.007
torsade / QT (curated PTs)	0.199%	19.3	**0/10**	**9/10**	**r = −0.81, p = 0.005**	r = −0.79, p = 0.006
torsade / QT (broad PTs)	0.659%	11.2	1/10	7/10	r = −0.76, p = 0.011	r = −0.72, p = 0.020
anaphylaxis	0.410%	5.2	1/4	1/4	n = 4, uninformative	n = 4, uninformative

The median marginal RR column orders the events by how drug-dominant they are: rhabdomyolysis 24.3, torsade 19.3 on the curated PT list and 11.2 on the broad one, anaphylaxis 5.2. That ordering is the paper's conditional made quantitative — the two events where both nulls misbehave are the two whose marginals are largest, and the arm that cannot test the claim is the one whose marginals are diffuse.

At $\Omega_{025} > 0$, Ω recovers **0/10** on torsade against 9/10 for the additive null — an apparently stronger version of the primary result on an independent drug-dominant event whose rate is within 4% of it. Amiodarone + sotalol, two of the most strongly QT-prolonging agents in use, scores $\Omega = -1.63$.

That apparent replication does not survive matched error rates, and we report it as a failed replication. Applying §4.3's analysis to torsade, the in-regime false-positive rates at $\Omega_{025} > 0$ are **2.0% for Ω and 42.8% for the additive null** — the additive null fires on nearly half of strongly-associated non-interacting pairs. Recalibrating both to a common in-regime rate (152 in-regime negatives):

operating point	additive	multiplicative
$\Omega_{025} > 0$ (as published)	9/10 @ 42.8%	0/10 @ 2.0%
matched at 5% in-regime FPR	**0/10**	**0/10**

operating point	additive	multiplicative
matched at 10%	**1/10**	**1/10**
matched at 20%	**4/10**	**3/10**

At any common error rate neither null recovers these pairs, and the additive null has no advantage. The 9-versus-0 result is an artefact of the conventional threshold sitting at wildly different error rates for the two measures on this event. What replicates on torsade is therefore the *asymmetry* — Ω again close to its nominal 2.5% (2.0%, on 152 in-regime negatives), the additive null far above it (42.8%), by a wider margin than on rhabdomyolysis — and not the recovery finding. On neither event is Ω's error rate the problem.

The marginal-strength gradient reproduces ($r = -0.81$, $p = 0.005$) and, as in §4.1, **it is not specific to the multiplicative null**: Ω_add shows $r = -0.79$ ($p = 0.006$) on the same pairs. The replication is of the *recovery failure*, which is unambiguous, and of the shared over-prediction — not of a gradient peculiar to Ω.

The anaphylaxis arm is invalid by construction, not underpowered, and we report it as a failed design rather than a weak result. It was intended to supply the negative case — an event with diffuse marginals where Ω should perform comparatively well. But anaphylaxis is overwhelmingly single-agent: there is no established drug pair whose *interaction* causes it. The pairs used are common co-exposures among agents that each cause anaphylaxis independently, which is a different thing entirely. **There is no interaction present for either null to detect, so no additional data would make this arm informative.** Two entries in the first version were worse than weak and have been removed: amoxicillin + clavulanate potassium is a fixed-dose combination product, not a drug–drug interaction at all, and contrast media + iohexol pairs a class with a member of that class. Four remain, and they establish nothing either way.

The calibration finding therefore generalises to a second event; the recovery advantage does not, and the condition has no valid negative case and remains a hypothesis. Constructing one requires an event with weak marginals *and* genuinely documented interacting pairs — the combination this study could not find.

5. Discussion

The pipeline recovers known pharmacology on the authors' own control set and has a characterised false-positive rate. What it does not do is provide evidence of enriching for genuine interactions beyond the drugs already nominated, and the central claim of an earlier version of this work — that temporal stability is a powerful discriminator — did not survive the validation it had not been given.

Two findings generalise beyond this event.

Neither null is usable at its conventional threshold when the drugs under study are the leading causes of the outcome. Against a nominal 2.5%, $\Omega_{025} > 0$ fires on 2.2% of strongly-associated non-interacting pairs and Ω_add,025 > 0 on 9.3%; on the replication event the additive figure is 42.8%. The recovery comparison that motivated this work — 4/16 against 12/16 — is largely an artefact of that miscalibration, falling to a one-to-two-pair difference at matched error rates and vanishing on the replication. This is precisely the situation for the best-characterised interaction classes, which is where a DDI method is most likely to be validated, so the failure mode is one a method could pass into routine use without encountering. The condition is checkable in advance: compute the marginal relative risks before choosing the null.

The mechanism is narrower than we first claimed. Observed joint event rates rise **far more shallowly** with marginal strength (r = +0.12, 95% CI −0.40 to +0.58) than either null predicts (r ≈ +0.94); joint risk saturates, and both models miss it. The multiplicative null overshoots much harder only because its expectation is larger at every point. We initially read the Ω-versus-marginals correlation as diagnostic of multiplicativity; it is not, since the additive null shows the same gradient (§4.1). Nor is the recovery comparison diagnostic unless the error rates are matched first — and once they are, the two nulls are close.

High-polypharmacy reports have leverage far out of proportion to their number. A tenth of a percent of cases supplied a third of all pair evidence at an enriched event rate. This is invisible from positive controls and only appears from the false-positive side.

A third observation is methodological rather than statistical: **an evaluation whose annotation is written by the same authors as the control set will show enrichment whether or not the method works.** The gap between 2.02× and 1.12× here is entirely that circularity. Screens of this kind reporting enrichment against author-curated reference lists should be read accordingly.

6. Limitations

- **The independent reference is FDA labelling, not a curated DDI database, and its coverage tracks US marketing status rather than pharmacology.** DrugBank is licensed and unavailable; the ONC list is available but is class-level and excludes the canonical myotoxicity pair (§2). Labelling is independent of the authors and captures 16/16 positive controls, but it is informed by FAERS itself, warns by class as well as by name, and is **absent entirely for 5.5% of screened ingredients — 9.9% of pairs**, among them fusidic acid. Widening the label reference to 800 ingredients raises the absent share to 17.2% and adds cerivastatin, the fibrates and telithromycin, but none of those four entered the top-200 screen and they do not bear on this result (§4.5). §4.5 reports the analysis restricted to label-covered pairs and the conclusion is unchanged, but the residual insensitivity biases enrichment *toward* this paper's own conclusion, not away from it. **A curated, ingredient-level, severity-graded DDI compendium remains the preferred reference and the primary required future work.** The measured false-positive rate is still an upper bound.

- **The screen's top-ranked pair is a documented interaction no reference contained.** Atorvastatin + fusidic acid (§4.7) has the highest event rate in the screen, is era-stable, and is contraindicated in practice, yet is absent from both the authors' list and openFDA. It is not a novel discovery, but its presence means the negative discovery result measures the references at least as much as the method, and the size of that effect is unknown.

- **The 16 positive controls were author-selected** and are the only evaluation set. Leave-one-out addresses optimism in the *estimand choice* only, not in the choice of controls. They are also **five victim drugs, not sixteen independent trials** — simvastatin is in seven — so recovery intervals are computed by resampling the victim drug and are correspondingly wide (50–100% on the powered subset). All 16 are now verified against FDA label text (§3.6); that verification had been intended and never performed.

- **The headline contrast is not robust to widening both design choices at once.** Of the four tier × role-policy arms (§4.3), three show the additive null recovering 7–8 more controls than the multiplicative one and the fourth shows no difference at all. The pre-specified arm is one of the three, but a reader is entitled to weigh the fourth.

- **Ingredient resolution accuracy is bounded, not directly measured.** FDA's own annotation of a given verbatim name is 98.99% self-consistent (§4.9), which bounds the level-2 backfill, but no manual audit against an external ground truth was performed.

- **The PT list had a single curator.** A second human reader was not available. A mechanical second annotation — every PT matching the myotoxicity vocabulary written independently for the label reference — flags 119 terms against the 23 curated, agreeing on 21. The mechanical pass is deliberately over-inclusive (it captures cardiomyopathy, fibromyalgia and muscle spasms, all excluded on clinical grounds), so the low Jaccard (0.174) reflects that over-inclusion rather than disagreement about the curated terms. This is a coverage check, **not** an inter-rater statistic, and does not substitute for a second reader.

- **Era bin boundaries are a researcher degree of freedom.** §4.9 shows the era-stable count runs from 84 to 3 depending purely on how many bins are demanded.

- **The near-duplicate rule uses exact set matching**, and provably misses the clusters documented in §4.4. The residual is now bounded: among event cases with a populated date and age, blocking on date/age/sex/country and comparing drug sets by Jaccard similarity finds 421 further pairs above 0.8 that the exact rule did not merge, affecting **226 cases (1.17% of event cases)**. A fuzzy rule was not adopted, since a Jaccard threshold is itself an unvalidated parameter, but the exposure is small.

- **No external validation on another reporting system, and the public interfaces cannot supply one.** VigiAccess, the free public window onto VigiBase, states that it "groups the search results both by active ingredient and geographically by continental region" and does not expose individual case records. It therefore returns single-drug adverse-reaction counts only and has **no facility for drug pairs**, so it cannot validate a drug-interaction result at all — only marginal associations. Pair-level VigiBase access requires a research agreement with the Uppsala Monitoring Centre; EudraVigilance likewise publishes single-substance report counts. Two pipeline *components* are externally benchmarked — deduplication against AEOLUS (8% apart) and ingredient resolution against DiAna (98.0% vs 98.94%) — and the central finding replicates on a second event within FAERS (§4.11). But no external reporting system has been used, and conclusions may be FAERS-specific.

- **Demographic subgroups are underpowered.** §4.9 reports sex-stratified results; both intervals include unity. Age and country were not stratified.

- **$\alpha = 0.5$ is corroborated by secondary sources, not by the primary.** Two independent sources state $\alpha = 0.5$ and the $\Omega_{025} > 0$ threshold (§2), and §4.8 shows no conclusion depends on α across a 20-fold range. The primary paper remains paywalled and unread by the authors.

- **Deduplication now benchmarked against AEOLUS** (Banda et al., 2016): over the identical window (Jan 2004 – Jun 2015) this pipeline retains 5,337,888 cases against their 4,928,413, a difference of 8%. AEOLUS applies a second pass deduplicating on event date, age, sex and country alone regardless of case number, which merges distinct patients sharing those fields, so a lower count is expected by construction. The two are consistent.

- **The in-regime error rates rest on constructed negative controls.** The purpose-built pool excludes documented interactions, so any undocumented true interaction it retains inflates the measured rate. **But it also excludes pairs whose two drugs are both on the implicated list — the configuration every positive control has — as too likely**

to be unrecorded interactions, and that exclusion removes the pairs most likely to fire, pushing the measured rate down. The two biases run in opposite directions and neither is quantified, so these rates are not a clean upper bound; the pool matches the positive controls on marginal strength but still not on implication status, and to that extent it inherits a weaker form of the defect it was built to remove. The matched-recovery table uses the 166 in-regime pairs of the generated pool and is noisy at the tails, and the in-regime cut is our choice rather than pre-specified.

- **Spontaneous reporting has no exposure denominator.** Nothing here estimates risk, only reporting disproportionality.
- **The negative result is bounded, not absolute.** At the widest screen it rests on 444 documented non-control pairs, with ~70% power at a true enrichment of 2.0 and ~30% at 1.5. It excludes enrichment above roughly 1.988× on the dependence-respecting interval. A real but modest enrichment would still be missed.
- **Screened drugs are selected on the outcome**, which is selection on the dependent variable. Reselecting by total report volume (§4.9) does not change the conclusion, but the negative controls are drawn from the same selected universe and share any induced bias.
- **The conditional claim is not fully established, and the intended negative case was invalid rather than underpowered.** The Ω failure replicates on torsade de pointes (§4.11), so it is not specific to rhabdomyolysis. But the *condition* — that failure follows from the drugs being the dominant cause — needs an event where marginals are weak and Ω performs well. The anaphylaxis arm was intended as that case and **cannot serve as one at any sample size**: anaphylaxis is essentially single-agent, so its "control" pairs are co-exposures among independently causative drugs rather than interacting pairs. There is nothing there for either null to detect. Constructing a valid negative case needs an event with weak marginals *and* documented interacting pairs, a combination we could not find.

7. Conclusion

A screen can determine in advance whether its choice of null is a real choice. Both statistics are the same shrunk log-ratio and differ only through the expectation, so wherever E_mult ≥ E_add the multiplicative signal set lies inside the additive one and the two nulls are one ordering at two thresholds. That ordering holds for 98.3% of drug-dominant pairs and 16.9% of diffuse ones, without a single exception among 13,208 ordered pairs, and it is computable from the marginals before any result is seen. Where it holds, a recovery comparison between the nulls measures the operating point and nothing else — which is what the eight-pair advantage in this study turns out to be, falling to one or two pairs at matched error rates and to none on the replication event.

The thresholds those comparisons should use are supplied. On a purpose-built pool of 2,345 strongly-associated non-interacting pairs — a construction the standard frequency-matched generator cannot produce — $\Omega_{025} > 0$ fires on 2.2% (95% CI 1.24–3.34%) and Ω_add,$_{025} > 0$ on 9.3% (7.29–11.32%) against a nominal 2.5%. Only the additive null is miscalibrated on error rate; Ω holds its advertised rate and is disqualified by power instead. Both figures are new: no prior work measures either null in the regime where DDI methods are validated. The temporal-stability filter, promoted in an earlier version of this work, does not survive validation against negative controls and is reported here as a negative result.

8. References

Verified against indexed sources. PMIDs and DOIs as listed.

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Secondary corroboration for the Ω definition (see §2): reference 12, a peer-reviewed review of DDI statistical methodologies, together with Uppsala Monitoring Centre operational documentation on drug–drug interaction signalling. The primary Norén et al. paper is paywalled and unread by the authors, so the definition used here rests on those secondary sources.

9. Computational environment

Single workstation, macOS on Apple silicon; Python 3.14.6, DuckDB 1.5.5, pandas 3.0.5, PyArrow 25.0.0, NumPy 2.5.1, SciPy 1.18.0, matplotlib. Exact pinned versions in `requirements.txt`. Download of the 90 archives takes roughly 30 minutes on a domestic connection; parsing 328M rows to Parquet about 3 minutes at 4 processes; the full analysis (`run_analysis`, `sensitivity`, `generalization`) about 12 minutes. Peak memory is bounded by a 10 GB DuckDB limit. No GPU. Total on-disk footprint 157 GB, of which 3.55 GB is the irreducible source archive set.

Data and code availability

The in-regime negative pool (`in_regime_pool.csv`, 19,826 pairs with the 2,345 in-regime ones flagged) and the per-drug marginal relative risks behind Table 2 (`rr_a`/`rr_b` in `tier_a_results.csv`) both ship, so the calibration rates and the mechanistic correlations can be recomputed without rebuilding the database.

All code, configuration and result tables are available at https://github.com/edanmn/faers-ddi-rhabdomyolysis. Every figure quoted in the Abstract and Results is generated into `results/canonical_numbers.json` by a single deterministic run and asserted against this text by `tests/test_canonical_numbers.py`; pipeline statistics quoted in Methods are persisted under `audit.provenance` in the same file and asserted alongside them. Figures drawn from cited work are attributed and not regenerated. The pipeline is deterministic: two full runs produce byte-identical output. `results/tables/download_manifest.csv` records a SHA-256 for each of the 90 archives. Per-phase development records, including errors made and corrected, are in `results/PHASE*_FINDINGS.md`.

This is research code and a research result. It is not clinical guidance.

Figures

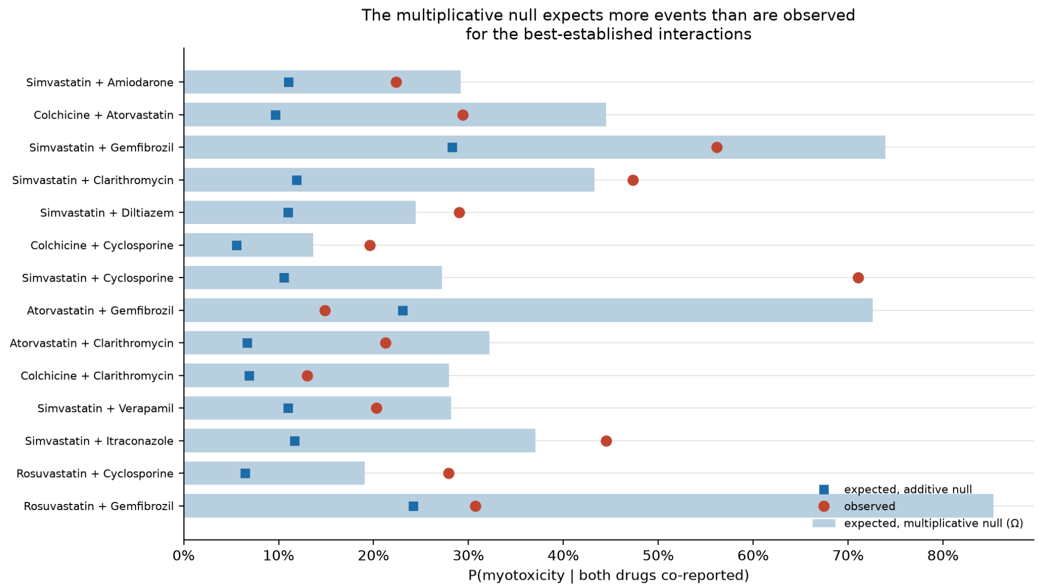

Figure 1: Observed proportion of co-reports carrying myotoxicity (red circles) against the proportion expected under the multiplicative null (bars) and the additive null (blue squares), for the 14 positive controls with $\geq$50 co-reports, ordered by co-report count. The multiplicative null expects more events than are observed for most established interactions, which is why Ω scores them as protective.

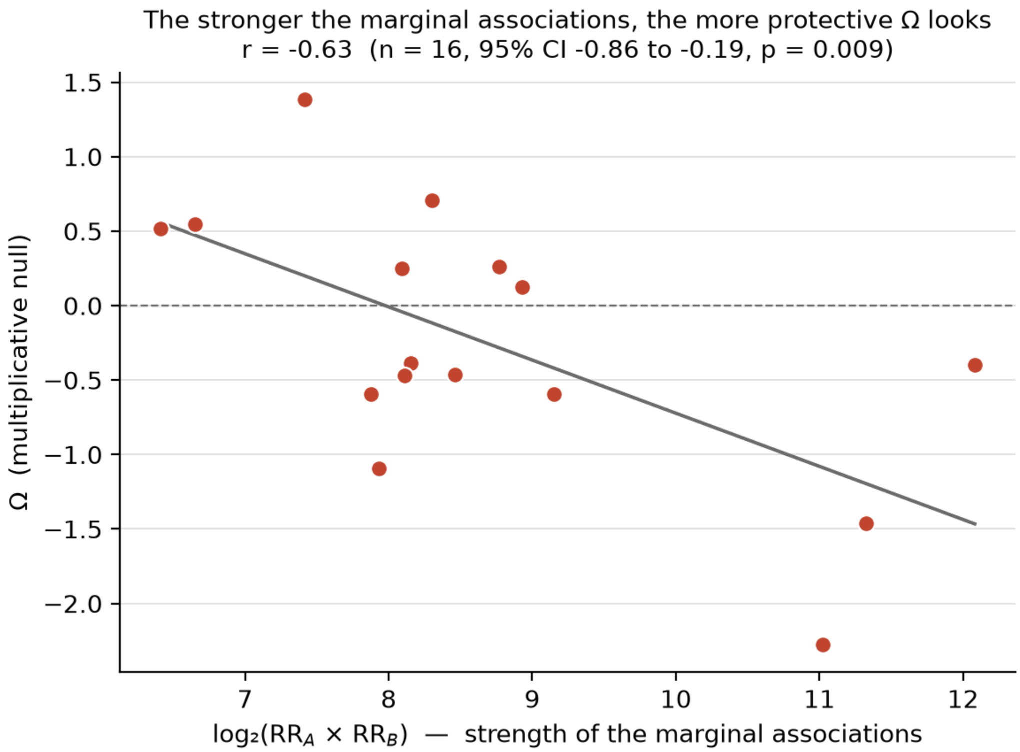

Figure 2: Ω for each positive control against $\log_2$(RR_A $\times$ RR_B), the combined strength of the two drugs' individual associations with the event. Line is ordinary least squares. The stronger the marginal associations, the more protective Ω appears.

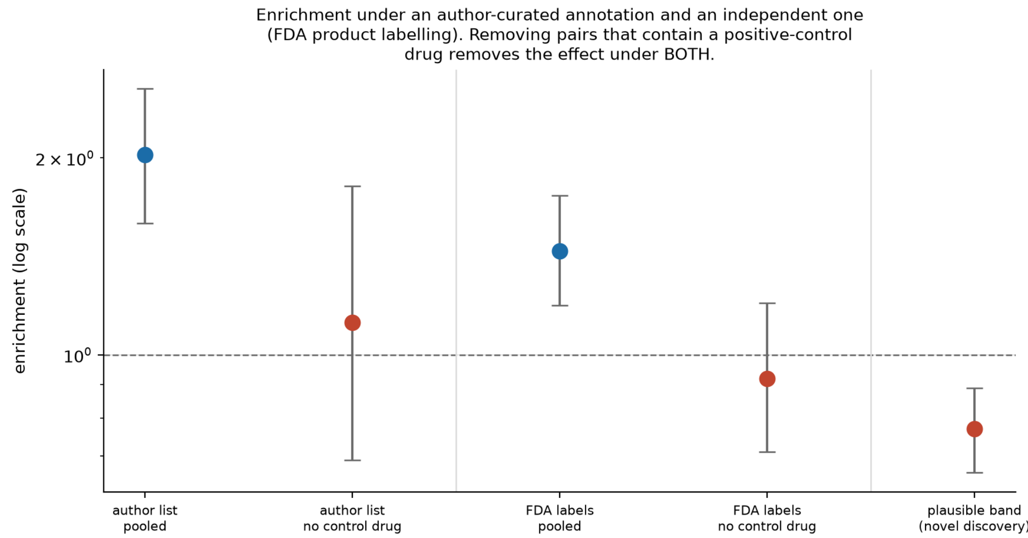

Figure 3: Signal enrichment relative to unsupported pairs, log scale, with 95% intervals. Five points in three groups. The first two are the author-curated annotation, pooled and then with pairs containing a positive-control drug removed; the middle two are the same contrast under the independent FDA-labelling annotation; the last is the plausible discovery band, the quantity those scopes are being compared against. Under both annotations, removing control-drug pairs moves enrichment to unity. The era-stable composition enrichment is deliberately **not** plotted here — it does not survive the two corrections applied to every other enrichment in this paper, and is tabulated with its corrections in §4.6.

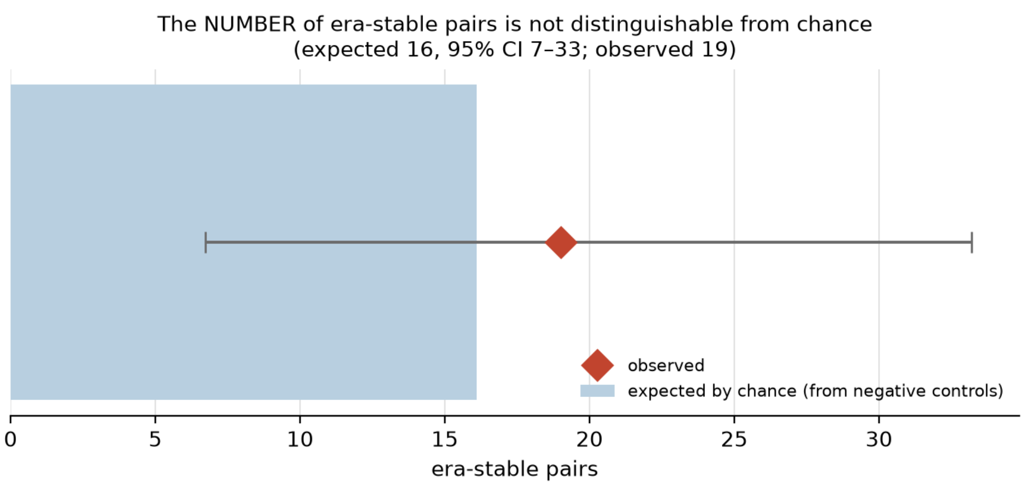

Figure 4: Number of era-stable pairs observed (red diamond) against the number expected by chance (bar, with 95% interval) computed by applying the same filter to the negative controls. The observed count lies inside the interval.

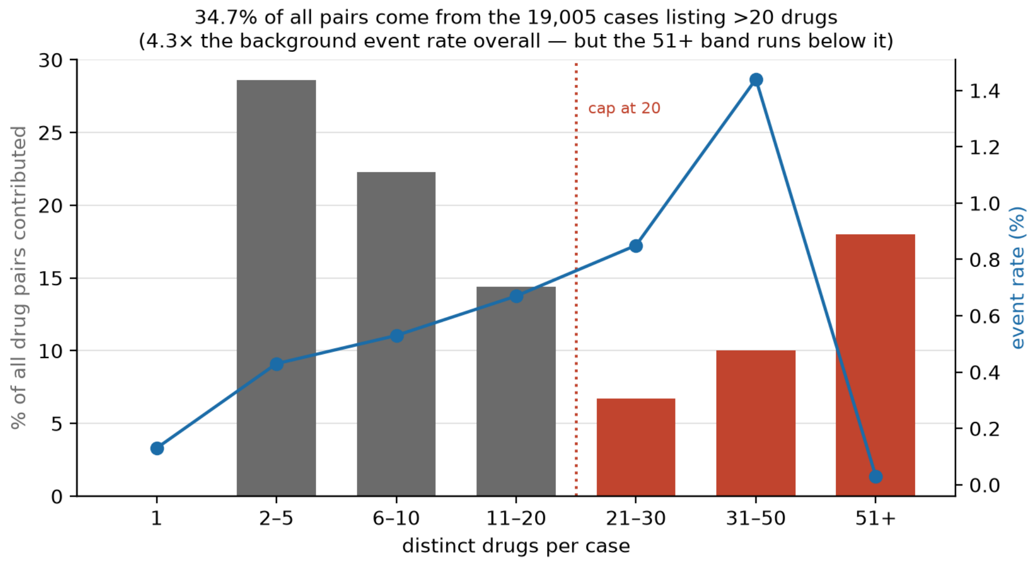

Figure 5: Percentage of all drug pairs contributed by cases in each drugs-per-case band (bars, left axis) and the myotoxicity event rate within that band (line, right axis). Dotted line marks the cap adopted at 20 drugs.

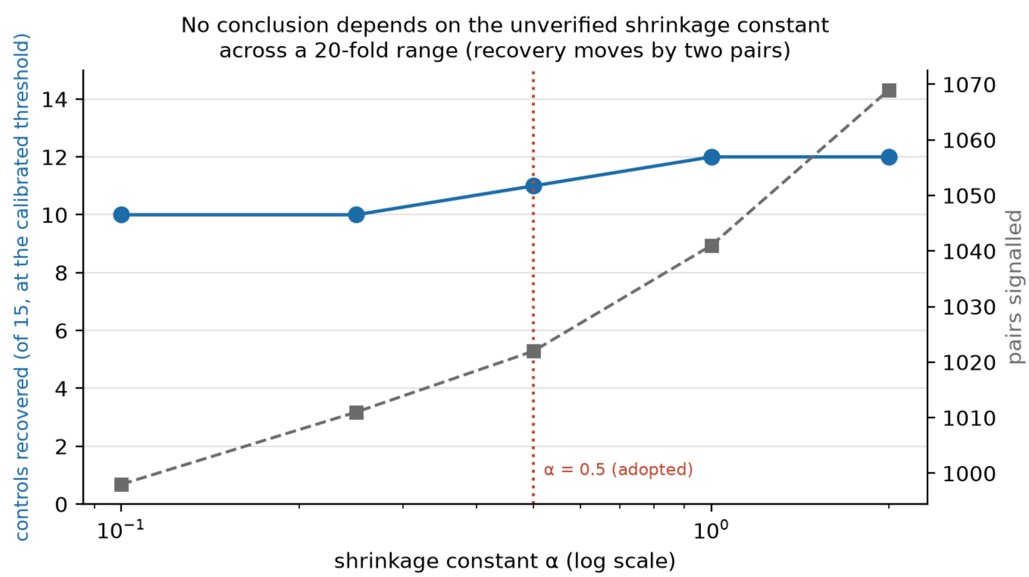

Figure 6: Positive controls recovered (left axis) and pairs signalled (right axis) as the shrinkage constant α varies over a 20-fold range, with the threshold recalibrated on the full negative pool at each value. Dotted line marks the adopted $\alpha = 0.5$.

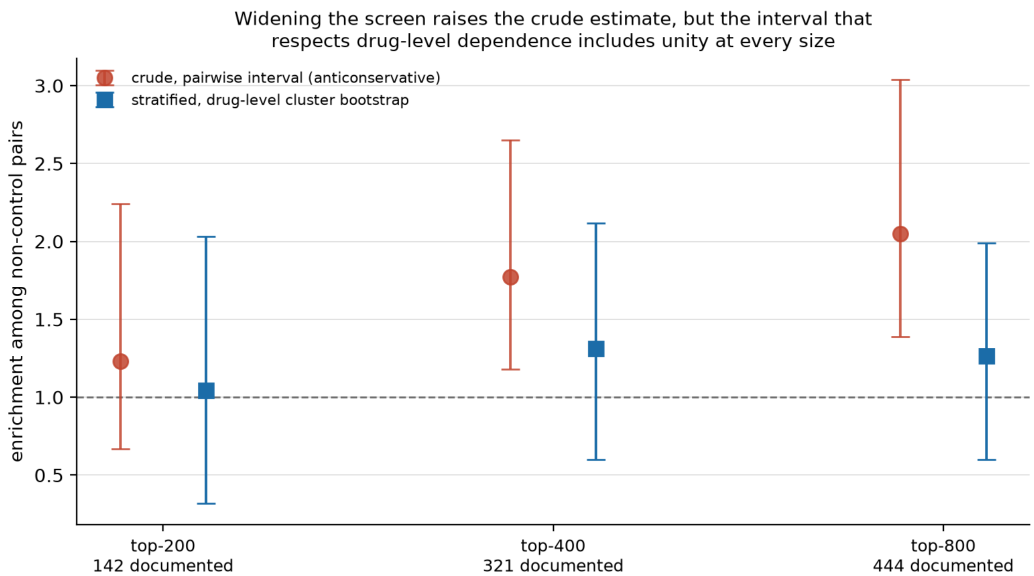

Figure 7: Enrichment among non-control pairs at three screen sizes, under the pairwise interval (red, anticonservative because pairs share drugs) and the drug-level cluster bootstrap (blue). The cluster interval includes unity at every size.